Reinforcement Learning Programming

CSCN8020 - Winter 2026 - Section 1

KAMAMO LESLEY

8984971

Assignment 2

Q-Learning on Taxi-v3 Environment

Introduction

This report presents the implementation and evaluation of Q-Learning on the Taxi-v3 environment. The agent was trained for 10,000 episodes under multiple learning rate (alpha) and exploration factor(epsilon) configurations. The primary metrics reported for the evaluation were the total number of steps taken per episode and the average return per episode. These metrics were used to assess convergence speed, policy efficiency, and overall learning performance across different learning rates and exploration parameter settings.

Talking Point 1

For every episode, the agent:
Observes the current state(**s**) → Chooses action using **ε-greedy** → Receives reward **r** → Updates the **q-value**

The update is the one that contains all the 3 key parameters, which control the different aspects of the agents' learning:

1. **Alpha α** (learning rate)
   It asks the question of "*How much do we trust new information compared to the old information?*"
   This controls how much we change the Q-value after each experience.

Talking Point 2

A large learning rate like **α = 0.2** makes updates big, which speeds up learning but can cause the values to overshoot and become unstable. A very small learning rate like **α = 0.001** produces tiny updates, leading to slow but steady learning. This slower pace usually requires many more episodes for the agent to converge.

2. **Gamma γ** (discount factor) → how much future reward matters
   Asks the questions – *"Am I planning for long-term reward or just quick wins?"*
   This controls how much future reward matters compared to the immediate reward.

Talking Point 3

When **γ is close to 0**, the agent focuses almost entirely on immediate rewards, leading to short-sighted, greedy behavior. When **γ is close to 1**, future rewards become important, so the agent plans ahead and behaves more strategically.

In the Taxi environment, successful navigation requires multi-step planning from pickup to drop-off, so a high γ (0.9) is used for good performance.

3. **Epsilon ε** (exploration rate) → how often we explore randomly
   Asks the question – *"Should I try something new or stick to what works?"*
   This controls the probability of random action.

Talking Point 4

A high exploration rate like **ε = 0.3** makes the agent try many different actions, which improves state coverage but slows convergence because it spends more time exploring than exploiting what it has learned. A low exploration rate like **ε = 0.1** pushes the agent toward exploitation, allowing it to converge faster but increasing the risk of getting stuck in a local optimum. Therefore, balancing ε is essential because too much exploration wastes time, while too little prevents the agent from discovering better long-term strategies.

Talking Point 5

Convergence occurs when the Q-values stop changing significantly and the agent's returns become stable with a flat, low-variance moving average. Also, when the number of steps per episode levels off, showing the agent can solve the task efficiently and consistently.

Observations and Analysis

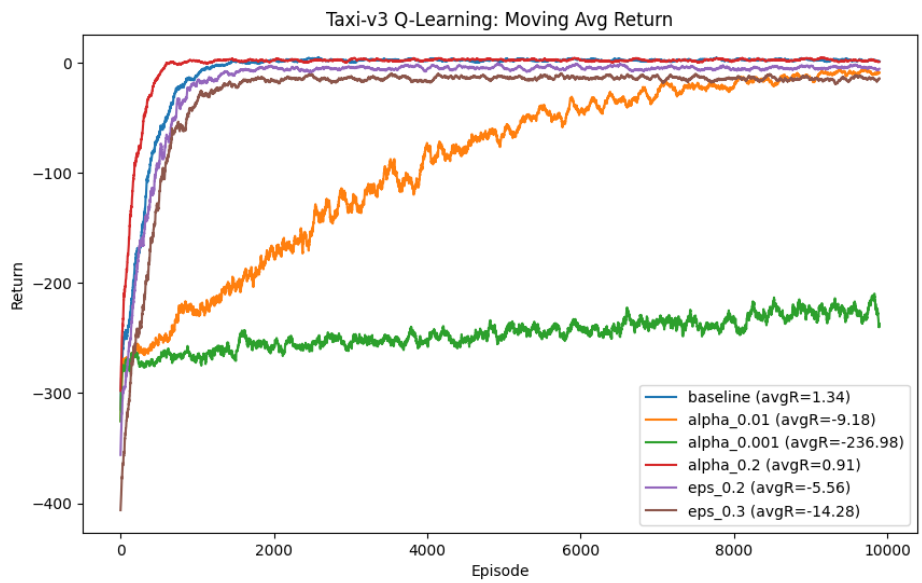
Analysis

Parameter Comparison Table

Config	Alpha	Epsilon	Avg Return (last100)	Avg Steps (last100)
Baseline	0.1	0.1	1.34	14.8

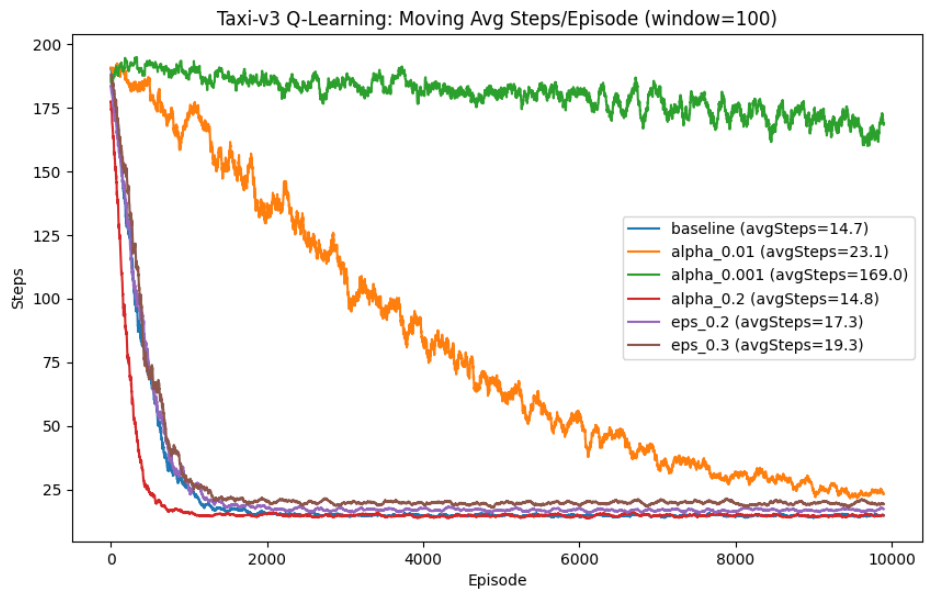
Alpha_0.01	0.01	0.1	-9.18	23.7
Alpha_0.001	0.001	0.1	-236.98	176.59
Alpha_0.2	0.2	0.1	0.91	15.23
Eps_0.2	0.1	0.2	-5.56	17.38
Eps_0.3	0.1	0.3	-14.28	19.53

Moving Average Return (Window=100)



This graph shows the moving average return of the Q-learning agent across 10,000 episodes for different learning rates (α) and exploration factors (ε). Moderate learning rates (α = 0.1 and α = 0.2) converge quickly to near-optimal returns, while very small learning rates (α = 0.001) result in extremely slow learning and poor overall performance. Higher exploration values (ε = 0.2 and 0.3) slow convergence and slightly reduce final performance because continued random exploration limits stable policy exploitation.

Moving Average Steps Plot



This graph shows the moving average number of steps per episode during training for different learning rates and exploration factors. Configurations with α = 0.1 and α = 0.2 quickly reduce the number of steps to around 14–15, indicating fast convergence to an efficient policy, while α = 0.001 remains very high (around 170 steps), showing poor learning. Higher exploration values (ε = 0.2 and 0.3) result in slightly more steps than the baseline because continued random exploration prevents the agent from consistently exploiting the optimal policy.

Talking Point 6

Lower steps per episode show that the agent is navigating efficiently and avoiding unnecessary movements. This pattern indicates that it has already learned a successful and reliable policy.

Observations

A very small learning rate (α = 0.001) resulted in poor convergence, extremely low returns, and a high number of steps per episode, indicating that the agent was learning too slowly due to minimal Q-value updates. In contrast, moderate learning rates (α = 0.1 and α = 0.2) converged much faster and achieved positive average returns, demonstrating more effective value updates. While a higher learning rate (α = 0.2) accelerated convergence, it may introduce slight instability in updates.

Overall, the best-performing configuration was the baseline ($\alpha = 0.1$, $\varepsilon = 0.1$), which achieved the highest average return (≈ 1.34) and maintained a low average number of steps (≈ 14.8), indicating efficient and stable learning.

Regarding exploration, increasing the exploration rate ($\varepsilon = 0.2$ and 0.3) improved early state-space coverage but reduced final performance due to continued random action selection. Higher exploration led to slower stabilization of returns and slightly more steps per episode, as the agent continued to take non-optimal actions even after learning a good policy.

Although increased exploration can be beneficial during early training, maintaining too high an exploration rate prevents consistent exploitation of the learned policy. Therefore, the baseline configuration ($\alpha = 0.1$, $\varepsilon = 0.1$) achieved the best balance between exploration and exploitation, resulting in stable convergence and strong overall performance.

Talking Point 7

The configuration of ($\alpha = 0.1$, $\varepsilon = 0.1$) works best because $\alpha = 0.2$ allows larger, faster updates while $\varepsilon = 0.1$ provides a strong balance between exploration and exploitation, leading to quick convergence, the highest average return, and consistently low step counts.

The learning curve remains stable because the agent learns efficiently without overshooting and explores just enough to avoid poor local optima, giving it the strongest overall performance among the tested settings.

Re-running Using Best Configuration

From your results (comparison table), the best-performing configuration is:

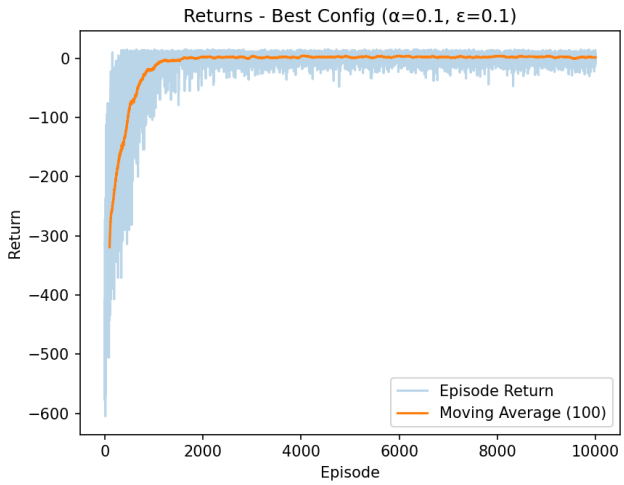
- Learning Rate **(α) = 0.1**
- Exploration Rate **(ε) = 0.1**
- Discount Factor **(γ) = 0.9** (unchanged)

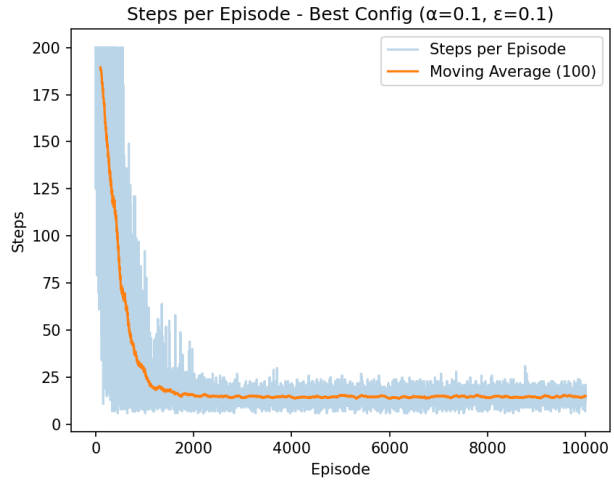
This configuration achieved:

- Highest average return **≈ 1.34**
- Low average steps **≈ 14.8**

This means fast and stable convergence.

Observation





The returns graph shows that the agent quickly improves from very low rewards and stabilizes near optimal performance after roughly the first 1,000–1,500 episodes, indicating successful convergence. Similarly, the steps-per-episode graph drops sharply from high initial values to around 14–15 steps and remains stable, confirming that the agent has learned an efficient and consistent policy.

Talking Point 8

With this configuration, training stabilizes much faster, with returns rising quickly. The steps per episode drop sharply and remain consistently low. The learning curves also show much lower variance than other settings, producing smoother performance and higher final returns.

Differences observed compared to other configurations

Parameter Change	Observed Effect
Smaller α (0.001)	Very slow updates which means no meaningful convergence
Smaller α (0.01)	Converges, but much slower
Larger α (0.2)	Slightly faster early learning but minor instability
Higher ε (0.2–0.3)	More exploration which means slower final stabilization